

DOACROSS cost-model accuracy — 2 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.1953	0.1854	+5.3%	5.3%
Short chain	20,000	1	1	2.2290	2.4325	-8.4%	8.4%
Short, four chains	20,000	4	1	1.1132	1.1514	-3.3%	3.3%
Short, 64 chains	20,000	64	1	1.1683	1.2587	-7.2%	7.2%
Long chain	200,000	1	1	19.3558	23.9546	-19.2%	19.2%
Long, four chains	200,000	4	1	9.9274	11.3475	-12.5%	12.5%
Long, 64 chains	200,000	64	1	9.9622	10.1433	-1.8%	1.8%
Compute in dependency	32,768	1	1	7.8926	7.1033	+11.1%	11.1%
Compute before wait	32,768	4	1	3.8016	3.5864	+6.0%	6.0%
Compute after post	32,768	4	1	3.7213	3.4653	+7.4%	7.4%
Two dependent streams	32,768	4	2	2.7001	3.5002	-22.9%	22.9%
Uneven lanes + compute	8,193	3	1	1.1662	1.1362	+2.6%	2.6%
Large arrays, 64 chains	1,000,000	64	1	50.1376	54.6414	-8.2%	8.2%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 3 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2447	0.2146	+14.1%	14.1%
Short chain	20,000	1	1	2.8041	2.7933	+0.4%	0.4%
Short, four chains	20,000	4	1	1.3744	1.3276	+3.5%	3.5%
Short, 64 chains	20,000	64	1	1.2179	1.0659	+14.3%	14.3%
Long chain	200,000	1	1	26.5829	27.0394	-1.7%	1.7%
Long, four chains	200,000	4	1	11.9670	11.7734	+1.6%	1.6%
Long, 64 chains	200,000	64	1	9.1616	8.2821	+10.6%	10.6%
Compute in dependency	32,768	1	1	9.7672	7.7242	+26.4%	26.4%
Compute before wait	32,768	4	1	3.1058	2.9433	+5.5%	5.5%
Compute after post	32,768	4	1	3.4095	3.0639	+11.3%	11.3%
Two dependent streams	32,768	4	2	2.8816	3.5793	-19.5%	19.5%
Uneven lanes + compute	8,193	3	1	0.8524	0.6574	+29.7%	29.7%
Large arrays, 64 chains	1,000,000	64	1	47.5148	42.8177	+11.0%	11.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3078	0.2633	+16.9%	16.9%
Short chain	20,000	1	1	3.6223	3.2448	+11.6%	11.6%
Short, four chains	20,000	4	1	0.9553	0.7932	+20.4%	20.4%
Short, 64 chains	20,000	64	1	1.0330	0.8285	+24.7%	24.7%
Long chain	200,000	1	1	28.5876	28.1206	+1.7%	1.7%
Long, four chains	200,000	4	1	8.5328	6.6481	+28.3%	28.3%
Long, 64 chains	200,000	64	1	8.1664	6.2459	+30.7%	30.7%
Compute in dependency	32,768	1	1	9.8205	7.4184	+32.4%	32.4%
Compute before wait	32,768	4	1	2.2552	1.8118	+24.5%	24.5%
Compute after post	32,768	4	1	2.3460	1.8815	+24.7%	24.7%
Two dependent streams	32,768	4	2	2.0420	2.0431	-0.1%	0.1%
Uneven lanes + compute	8,193	3	1	0.9783	0.7463	+31.1%	31.1%
Large arrays, 64 chains	1,000,000	64	1	39.7230	29.7494	+33.5%	33.5%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 5 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4470	0.3605	+24.0%	24.0%
Short chain	20,000	1	1	3.4409	3.2821	+4.8%	4.8%
Short, four chains	20,000	4	1	1.2935	1.1121	+16.3%	16.3%
Short, 64 chains	20,000	64	1	1.1668	0.8810	+32.4%	32.4%
Long chain	200,000	1	1	30.9459	29.0019	+6.7%	6.7%
Long, four chains	200,000	4	1	8.9297	7.7739	+14.9%	14.9%
Long, 64 chains	200,000	64	1	7.9751	6.0425	+32.0%	32.0%
Compute in dependency	32,768	1	1	10.6832	8.0147	+33.3%	33.3%
Compute before wait	32,768	4	1	2.5884	2.1753	+19.0%	19.0%
Compute after post	32,768	4	1	2.5667	2.0373	+26.0%	26.0%
Two dependent streams	32,768	4	2	2.5642	2.5033	+2.4%	2.4%
Uneven lanes + compute	8,193	3	1	0.9845	0.7299	+34.9%	34.9%
Large arrays, 64 chains	1,000,000	64	1	40.5817	30.3831	+33.6%	33.6%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 6 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.5149	0.4654	+10.6%	10.6%
Short chain	20,000	1	1	3.9939	3.6206	+10.3%	10.3%
Short, four chains	20,000	4	1	1.2455	0.9662	+28.9%	28.9%
Short, 64 chains	20,000	64	1	1.1116	0.8492	+30.9%	30.9%
Long chain	200,000	1	1	33.1028	31.8784	+3.8%	3.8%
Long, four chains	200,000	4	1	8.1871	6.8967	+18.7%	18.7%
Long, 64 chains	200,000	64	1	7.3276	5.4836	+33.6%	33.6%
Compute in dependency	32,768	1	1	11.4949	8.3315	+38.0%	38.0%
Compute before wait	32,768	4	1	2.4750	2.0024	+23.6%	23.6%
Compute after post	32,768	4	1	2.2371	1.8472	+21.1%	21.1%
Two dependent streams	32,768	4	2	2.2750	2.1554	+5.5%	5.5%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	37.2661	27.3762	+36.1%	36.1%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 7 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.5973	0.4900	+21.9%	21.9%
Short chain	20,000	1	1	4.5796	4.3431	+5.4%	5.4%
Short, four chains	20,000	4	1	1.5191	1.3342	+13.9%	13.9%
Short, 64 chains	20,000	64	1	1.3477	0.9681	+39.2%	39.2%
Long chain	200,000	1	1	37.1864	34.2498	+8.6%	8.6%
Long, four chains	200,000	4	1	11.4051	9.3825	+21.6%	21.6%
Long, 64 chains	200,000	64	1	7.7432	5.3415	+45.0%	45.0%
Compute in dependency	32,768	1	1	11.8220	9.1342	+29.4%	29.4%
Compute before wait	32,768	4	1	2.3490	1.7855	+31.6%	31.6%
Compute after post	32,768	4	1	2.5577	2.0840	+22.7%	22.7%
Two dependent streams	32,768	4	2	2.6539	2.4692	+7.5%	7.5%
Uneven lanes + compute	8,193	3	1	1.2941	0.9606	+34.7%	34.7%
Large arrays, 64 chains	1,000,000	64	1	38.5430	25.9047	+48.8%	48.8%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 8 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.6815	0.6047	+12.7%	12.7%
Short chain	20,000	1	1	4.6078	4.4820	+2.8%	2.8%
Short, four chains	20,000	4	1	1.3079	1.0805	+21.1%	21.1%
Short, 64 chains	20,000	64	1	0.9610	0.7796	+23.3%	23.3%
Long chain	200,000	1	1	39.4293	35.7823	+10.2%	10.2%
Long, four chains	200,000	4	1	9.5309	7.1289	+33.7%	33.7%
Long, 64 chains	200,000	64	1	7.0871	4.2631	+66.2%	66.2%
Compute in dependency	32,768	1	1	12.2134	9.1156	+34.0%	34.0%
Compute before wait	32,768	4	1	2.3355	1.7261	+35.3%	35.3%
Compute after post	32,768	4	1	2.2367	1.6321	+37.0%	37.0%
Two dependent streams	32,768	4	2	2.5859	2.2954	+12.7%	12.7%
Uneven lanes + compute	8,193	3	1	1.5258	1.0790	+41.4%	41.4%
Large arrays, 64 chains	1,000,000	64	1	33.7112	20.0206	+68.4%	68.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 2 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2009	0.1854	+8.4%	8.4%
Short chain	20,000	1	1	3.8776	2.4325	+59.4%	59.4%
Short, four chains	20,000	4	1	3.9035	1.1514	+239.0%	239.0%
Short, 64 chains	20,000	64	1	3.9053	1.2587	+210.3%	210.3%
Long chain	200,000	1	1	37.5732	23.9546	+56.9%	56.9%
Long, four chains	200,000	4	1	37.4299	11.3475	+229.9%	229.9%
Long, 64 chains	200,000	64	1	37.2829	10.1433	+267.6%	267.6%
Compute in dependency	32,768	1	1	9.1535	7.1033	+28.9%	28.9%
Compute before wait	32,768	4	1	8.2698	3.5864	+130.6%	130.6%
Compute after post	32,768	4	1	8.1655	3.4653	+135.6%	135.6%
Two dependent streams	32,768	4	2	11.7668	3.5002	+236.2%	236.2%
Uneven lanes + compute	8,193	3	1	2.0967	1.1362	+84.5%	84.5%
Large arrays, 64 chains	1,000,000	64	1	187.4033	54.6414	+243.0%	243.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 3 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2088	0.2146	-2.7%	2.7%
Short chain	20,000	1	1	2.8005	2.7933	+0.3%	0.3%
Short, four chains	20,000	4	1	2.8454	1.3276	+114.3%	114.3%
Short, 64 chains	20,000	64	1	2.8385	1.0659	+166.3%	166.3%
Long chain	200,000	1	1	26.7206	27.0394	-1.2%	1.2%
Long, four chains	200,000	4	1	26.6356	11.7734	+126.2%	126.2%
Long, 64 chains	200,000	64	1	26.2402	8.2821	+216.8%	216.8%
Compute in dependency	32,768	1	1	7.6679	7.7242	-0.7%	0.7%
Compute before wait	32,768	4	1	5.6305	2.9433	+91.3%	91.3%
Compute after post	32,768	4	1	5.8570	3.0639	+91.2%	91.2%
Two dependent streams	32,768	4	2	8.1330	3.5793	+127.2%	127.2%
Uneven lanes + compute	8,193	3	1	1.5949	0.6574	+142.6%	142.6%
Large arrays, 64 chains	1,000,000	64	1	132.7389	42.8177	+210.0%	210.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2419	0.2633	-8.1%	8.1%
Short chain	20,000	1	1	2.3514	3.2448	-27.5%	27.5%
Short, four chains	20,000	4	1	2.2874	0.7932	+188.4%	188.4%
Short, 64 chains	20,000	64	1	2.3318	0.8285	+181.4%	181.4%
Long chain	200,000	1	1	20.9415	28.1206	-25.5%	25.5%
Long, four chains	200,000	4	1	21.1777	6.6481	+218.6%	218.6%
Long, 64 chains	200,000	64	1	20.8213	6.2459	+233.4%	233.4%
Compute in dependency	32,768	1	1	6.5176	7.4184	-12.1%	12.1%
Compute before wait	32,768	4	1	4.4809	1.8118	+147.3%	147.3%
Compute after post	32,768	4	1	4.5053	1.8815	+139.4%	139.4%
Two dependent streams	32,768	4	2	6.4106	2.0431	+213.8%	213.8%
Uneven lanes + compute	8,193	3	1	1.3861	0.7463	+85.7%	85.7%
Large arrays, 64 chains	1,000,000	64	1	103.6812	29.7494	+248.5%	248.5%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 5 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3091	0.3605	-14.3%	14.3%
Short chain	20,000	1	1	2.0272	3.2821	-38.2%	38.2%
Short, four chains	20,000	4	1	2.0422	1.1121	+83.6%	83.6%
Short, 64 chains	20,000	64	1	2.0558	0.8810	+133.3%	133.3%
Long chain	200,000	1	1	17.5829	29.0019	-39.4%	39.4%
Long, four chains	200,000	4	1	17.5488	7.7739	+125.7%	125.7%
Long, 64 chains	200,000	64	1	17.7528	6.0425	+193.8%	193.8%
Compute in dependency	32,768	1	1	6.1591	8.0147	-23.2%	23.2%
Compute before wait	32,768	4	1	3.9512	2.1753	+81.6%	81.6%
Compute after post	32,768	4	1	3.8453	2.0373	+88.7%	88.7%
Two dependent streams	32,768	4	2	5.4144	2.5033	+116.3%	116.3%
Uneven lanes + compute	8,193	3	1	1.2380	0.7299	+69.6%	69.6%
Large arrays, 64 chains	1,000,000	64	1	88.5795	30.3831	+191.5%	191.5%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 6 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3529	0.4654	-24.2%	24.2%
Short chain	20,000	1	1	1.8210	3.6206	-49.7%	49.7%
Short, four chains	20,000	4	1	1.8564	0.9662	+92.1%	92.1%
Short, 64 chains	20,000	64	1	1.8354	0.8492	+116.1%	116.1%
Long chain	200,000	1	1	15.4578	31.8784	-51.5%	51.5%
Long, four chains	200,000	4	1	15.4563	6.8967	+124.1%	124.1%
Long, 64 chains	200,000	64	1	15.4789	5.4836	+182.3%	182.3%
Compute in dependency	32,768	1	1	5.7775	8.3315	-30.7%	30.7%
Compute before wait	32,768	4	1	3.5287	2.0024	+76.2%	76.2%
Compute after post	32,768	4	1	3.4070	1.8472	+84.4%	84.4%
Two dependent streams	32,768	4	2	4.6699	2.1554	+116.7%	116.7%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	77.7185	27.3762	+183.9%	183.9%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 7 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4119	0.4900	-15.9%	15.9%
Short chain	20,000	1	1	1.7731	4.3431	-59.2%	59.2%
Short, four chains	20,000	4	1	1.6978	1.3342	+27.3%	27.3%
Short, 64 chains	20,000	64	1	1.8079	0.9681	+86.8%	86.8%
Long chain	200,000	1	1	14.0566	34.2498	-59.0%	59.0%
Long, four chains	200,000	4	1	14.2448	9.3825	+51.8%	51.8%
Long, 64 chains	200,000	64	1	14.2117	5.3415	+166.1%	166.1%
Compute in dependency	32,768	1	1	5.7505	9.1342	-37.0%	37.0%
Compute before wait	32,768	4	1	3.0780	1.7855	+72.4%	72.4%
Compute after post	32,768	4	1	3.2260	2.0840	+54.8%	54.8%
Two dependent streams	32,768	4	2	4.2502	2.4692	+72.1%	72.1%
Uneven lanes + compute	8,193	3	1	1.1826	0.9606	+23.1%	23.1%
Large arrays, 64 chains	1,000,000	64	1	69.7890	25.9047	+169.4%	169.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 8 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4487	0.6047	-25.8%	25.8%
Short chain	20,000	1	1	1.6842	4.4820	-62.4%	62.4%
Short, four chains	20,000	4	1	1.6432	1.0805	+52.1%	52.1%
Short, 64 chains	20,000	64	1	1.6275	0.7796	+108.8%	108.8%
Long chain	200,000	1	1	12.7944	35.7823	-64.2%	64.2%
Long, four chains	200,000	4	1	13.2272	7.1289	+85.5%	85.5%
Long, 64 chains	200,000	64	1	13.1234	4.2631	+207.8%	207.8%
Compute in dependency	32,768	1	1	5.5842	9.1156	-38.7%	38.7%
Compute before wait	32,768	4	1	3.0315	1.7261	+75.6%	75.6%
Compute after post	32,768	4	1	3.0415	1.6321	+86.4%	86.4%
Two dependent streams	32,768	4	2	3.9358	2.2954	+71.5%	71.5%
Uneven lanes + compute	8,193	3	1	1.2700	1.0790	+17.7%	17.7%
Large arrays, 64 chains	1,000,000	64	1	63.8355	20.0206	+218.8%	218.8%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before holdout. Model: per-lane synchronization; width-specific batch profiles.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.